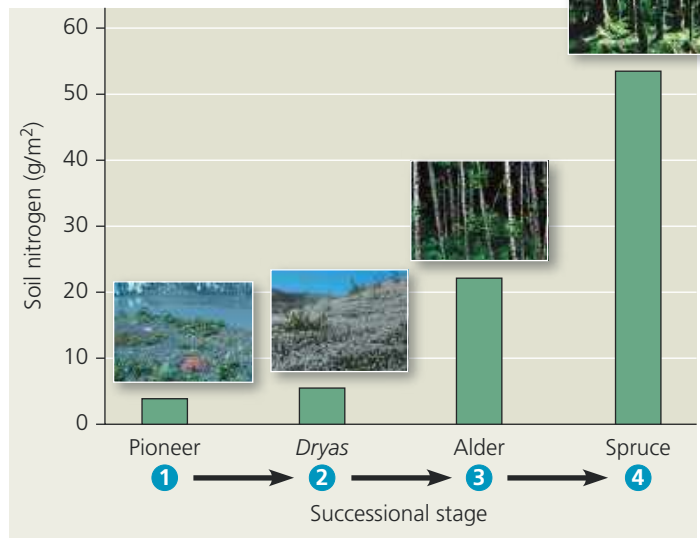


▼ **Figure 54.27** Changes in soil nitrogen content during succession at Glacier Bay.



MAKE CONNECTIONS Figure 37.11 illustrates two types of atmospheric nitrogen fixation by prokaryotes. At the earliest stages of primary succession, before any plants are present at a site, which type of nitrogen fixation would occur, and why?

species can facilitate colonization by new plant species during succession.

In contrast to primary succession, **secondary succession** involves the recolonization of an area after a major disturbance has removed most but not all of the organisms in a community, as in Yellowstone following the 1988 fires (see Figure 54.25). Following the disturbance, the area may return to something like its original state. For instance, in a forested area that has been cleared for farming and later abandoned, the earliest plants to recolonize are often herbaceous species that grow from windblown or animal-borne seeds. If the area has not been burned or heavily grazed, woody shrubs may in time replace most of the herbaceous species, and forest trees may eventually replace most of the shrubs.

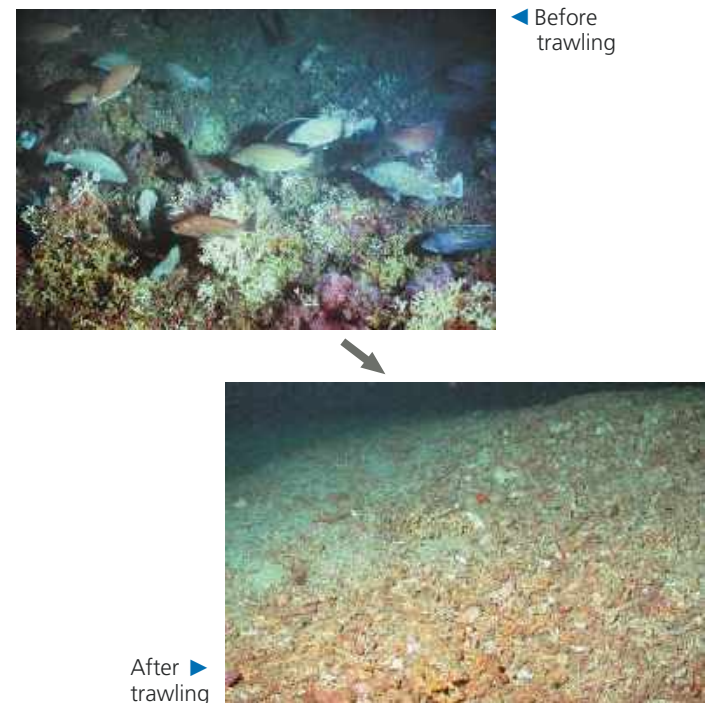
Human Disturbance

Ecological succession is a response to disturbance of the environment, and some of the strongest disturbances result from human activities. Agricultural development has disrupted what were once the vast grasslands of the North American prairie. Tropical rain forests are quickly disappearing as a result of clear-cutting for lumber, cattle grazing, and farmland. Centuries of overgrazing and agricultural disturbance have contributed to famine in parts of Africa by turning seasonal grasslands into vast barren areas.

Human actions have disturbed marine ecosystems as well as terrestrial ones. The effects of ocean trawling, in which boats drag weighted nets across the seafloor, are similar to those of clear-cutting a forest or plowing a field. The trawls scrape and scour corals and other life on the seafloor (Figure 54.28). In a typical

▼ **Figure 54.28** Disturbance of the ocean floor by trawling.

These photos show the seafloor off northwestern Australia before and after deep-sea trawlers have passed through.



year, ships trawl an area about the size of South America, 150 times larger than the area of forests that are clear-cut annually.

Because disturbance by human activities is often severe, it reduces species diversity in many communities. In Chapter 56, we'll take a closer look at how human-caused disturbance is affecting the diversity of life.

CONCEPT CHECK 54.3

1. Why do high and low levels of disturbance usually reduce species diversity? Why does an intermediate level of disturbance promote species diversity?
2. During succession, how might the early species facilitate the arrival of other species?
3. **WHAT IF?** Most prairies experience regular fires, typically every few years. If these disturbances were relatively modest, how would the species diversity of a prairie likely be affected if no burning occurred for 100 years? Explain.

For suggested answers, see Appendix A.

CONCEPT 54.4

Biogeographic factors affect community diversity

So far, we have examined relatively small-scale or local factors that influence the diversity of communities, including the effects of species interactions, foundation species, and many types of disturbances. Large-scale biogeographic factors also contribute to the tremendous range of diversity observed in biological communities.